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ORIGINAL ARTICLE

Dynamic asset allocation strategy: an economic regime approach

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Abstract
This paper presents a practical investment framework for dynamic asset allocation strategies based on changes in the macro-environment. To identify economic regimes, we use macro-indicators that track monthly growth and inflation of the US economy. We then demonstrate that the regimes divided by changes in growth and inflation trends successfully partition the historical performance of asset classes, and construct a regime-based dynamic strategy for shifting exposures toward attractive assets according to economic regimes. Out-of-sample analysis suggests that the dynamic approach outperforms the static approach after accounting for transaction costs, leading to a higher risk-adjusted return and information ratio. These results have crucial implications for portfolio managers seeking to develop a dynamic asset allocation strategy throughout economic cycles to enhance long-term portfolio performance.

Keywords Dynamic asset allocation strategy · Economic regimes · l_1 trend filtering

JEL Classification G11 · vG15 · E32

Introduction

It is well known that asset class performance is strongly linked to economic conditions (e.g., Fama and French 1989; Ferson and Harvey 1991; Lustig and Verdelhan 2012; Ilmanen et al. 2014). Generally, risky assets such as stocks outperform other assets during expansions, whereas low-risk assets including government bonds offer the best returns during recessions. The dynamic nature of financial markets throughout the business cycle presents a major challenge for static portfolios such as 60/40 in stocks and bonds.

This paper proposes a practical investment framework for a dynamic asset allocation strategy based on changes in economic regimes and investigates whether the regime-based dynamic approach produces better long-term results than the static approach. We begin by constructing monthly growth and inflation indicators that track the US economic activity and inflation, respectively. That is, we adopt a now-casting approach that has been extensively used in applied macroeconomic work to ameliorate the modeling of real-time business cycles, as described by Beber et al. (2015) and Bok et al. (2017). We then demonstrate that increases and decreases in growth and inflation trends clearly separate the historical performances of different asset classes. These time-varying risk and return properties throughout economic regimes can be exploited to enhance portfolio performance by tilting exposures toward effective assets according to the different regimes.

To assess the out-of-sample performance of the regime-based dynamic strategy, we first construct a risk parity portfolio as a benchmark for performance comparison. Following the methodology developed by Haesen et al. (2017) and Jurczenko and Teiletche (2018), we construct a dynamic portfolio by incorporating an investor's regime-based view of expected returns into the benchmark. We then conduct an out-of-sample experiment to compare the performance of the dynamic portfolio to that of the static benchmark. The key finding is that the regime-based dynamic strategy significantly outperforms the benchmark after accounting for transaction costs, resulting in a higher risk-adjusted return

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and information ratio. In particular, the increase in the risk-adjusted return and information ratio is driven largely by a significant increase in absolute returns. This result has noteworthy implications for portfolio managers seeking to outperform the benchmark and improve long-term portfolio performance.

This study extends the literature on regime-based asset allocation strategies in several ways. First, in contrast to previous studies that heavily rely on a Markov-switching model to detect market regimes using asset returns directly, such as Ang and Bekaert (2004), Bulla et al. (2011), Kritzman et al. (2012), Nystrup et al. (2015), and Nystrup et al. (2018), this paper adopts a more fundamental approach by identifying economic regimes that are more persistent and stable than market regimes. By doing so, we can avoid the major problems of the practical applicability of the Markov-switching model, including the high sensitivity of parameter estimation and regime changes to out-of-sample analysis. Second, we build useful indicators tracking the US economic activity and inflation and show that it successfully captures the well-documented properties of the US business cycle. Third, we present a practical and novel methodology to identify economic regimes by applying a trend filtering method to the growth and inflation indicators and extracting momentum changes in their trends. This method is a simple model-free approach that is broadly robust to parameter estimation uncertainty and can be easily implemented by practitioners. Finally, several studies, such as Vliet and Blitz (2011), Sheikh and Sun (2012), and Ilmanen et al. (2014), similarly emphasize the importance of changes in the macro-environment for dynamic asset allocation strategies, but they mainly focus on in-sample analysis. However, this paper contains out-of-sample analysis to firmly test the profitability of a regime-based dynamic strategy.

The remainder of this paper is organized as follows. "Constructing macro-indicators" section describes how to construct the growth and inflation indicators and "Identifying economic regimes" section discusses the methodology to identify economic regimes. "Asset class performance across economic regimes" section presents descriptive statistics for different asset classes across economic regimes. "Out-of-sample analysis" section reports the empirical results of out-of-sample performance. "Sensitivity analysis" section checks the sensitivity of the results and "Conclusion" section presents concluding remarks.

Constructing macro-indicators

To characterize economic regimes in a timely manner, we construct a nowcasting indicator that has been widely used by central banks and investors and proven to be extremely effective for tracking current economic conditions and identifying their implications for financial markets (e.g., Beber et al. 2015; Bok et al. 2017). Specifically, we use two separate indicators that track monthly growth and inflation of the US economy, respectively. This choice is motivated by the well-documented fact that growth and inflation are the most critical macroeconomic factors affecting asset returns.

Growth indicator

When constructing a growth indicator, we must note the limitations associated with macroeconomic data. Typically, economists rely on macro-variables, such as gross domestic product or industrial production, to analyze economic conditions. However, these measures are of little practical value for investment decisions because they are released with a significant lag and are subject to multiple revisions. More importantly, financial markets tend to lead economic activities. Therefore, by the time these macro-data are available, they only reflect information about the past, providing scarce information for predicting future asset returns.

To overcome these drawbacks, similar to Vliet and Blitz (2011) and Kwon (2022), we construct a growth indicator that consists of five well-known leading variables: Yield spread between the 10 year treasury yield and effective federal fund rate, credit spread between the Moody's Baa corporate bond yield and 10 year treasury yield, four-week moving average of initial jobless claims, total units of building permits, and CBOE Volatility Index (VIX).[1] All data are monthly and can be easily obtained from the Federal Reserve Economic Data maintained by the St. Louis Fed. The sample period covers the period from January 1976 to December 2020.[2]

The selection of the variables is based on the well-documented evidence. As argued by Estrella and Hardouvelis (1991) and Ang et al. (2006), the yield spread is a principal leading indicator for predicting economic slowdowns with an inverted yield curve signaling a deceleration in output. As highlighted by Gilchrist et al. (2009) and Gilchrist and Zakrajšek (2012), the credit spread is also considered a crucial indicator of the financial health of the economy, with widened credit spreads implying a deterioration in the overall business environment. Initial jobless claims, which measure the number of people filing new claims for

[1] A growth indicator may be defined by using different inputs, but we find that these variables have statistically significant predictive abilities for future economic activity and are found to capture the US business cycle successfully.

[2] The starting date of the sample period is determined by the availability of Barclays US Aggregate Bond Index, which will be utilized in our empirical work. Because the VIX index is not available until 1990, following Bloom (2009), we compute the monthly standard deviation of the daily S &P500 index as a proxy.

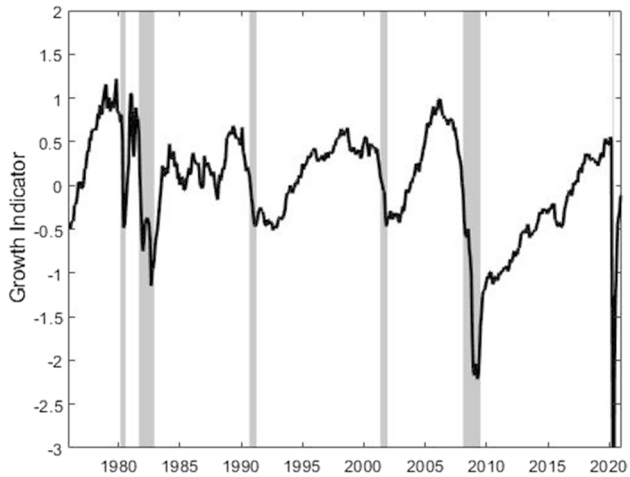


Fig. 1 Growth indicator

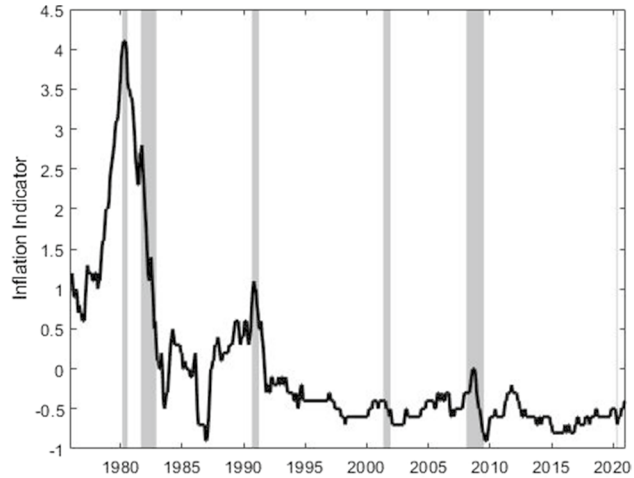


Fig. 2 Inflation indicator

unemployment insurance benefits, are one of the most sensitive and widely used leading indicators to reflect a sudden change in labor markets (e.g., Gavin and Kliesen 2002; Kliesen and Wheelock 2012). Additionally, building permits that measure new privately owned housing units authorized by permit-issuing places are commonly used as leading indicators, with their inversion typically serving as a critical signal of upcoming recessions (e.g., Berge 2015; Liu and Moench 2016). This is because the housing industry is one of the most cyclical sectors in the US economy and accounts for more than 25% of total investment spending. Finally, stock market volatility, which reflects investor sentiment and is one proxy for uncertainty, has been known to be intimately linked to macroeconomic fluctuations, as highlighted by Bloom (2009). Increased uncertainty generally has a strong and negative effect on economic activity, reducing consumption and investment.

We obtain each variable available at the end of each month and normalize it to have zero mean and unit variance.[3] To acquire the main factor summarizing variations in the different variables, we conduct principle component analysis and extract the first principal component of the five normalized variables, which is considered as the growth indicator. Figure 1 plots our growth indicator over the last 45 years with the National Bureau of Economic Research (NBER) recession periods denoted by shaded areas. Every economic cycle differs in duration and persistence, but has certain repetitive patterns over time with expansions followed by contractions. In other words, economic conditions always return to their potential level and every acceleration phase is inevitably followed by deceleration and vice versa. Particularly, the falling growth indicator after peak captures almost all official NBER recession periods reasonably well.

Inflation indicator

Traditionally, inflation has also been an important factor that has a significant impact on asset returns (e.g., Fama and Schwert 1977; Marshall 1992). To construct our inflation indicator, we rely on the underlying inflation gauge (UIG) developed by the New York Fed to track the US inflation monthly. The UIG measure represents inflation trends and is defined as the persistent part of the common component of various economic and financial series.[4] It is well known that the UIG measure not only outperforms core inflation measures in tests of forecast accuracy, but also provides more timely and precise signals for turning points in inflation. Figure 2 illustrates our inflation indicator for the past 45 years with the NBER recession periods denoted by shaded areas.[5] As widely documented in the literature, the US inflation has stabilized greatly in the 1990s until recently, but has spiked up post-pandemic. Additionally, it exhibits certain repetitive features over time with a peak before a recession, followed by a sharp decline.

[3] All variables, except for building permits, are inverted by imposing a negative sign on them due to their countercyclical properties. Additionally, because initial jobless claims and building permits are available with a one month delay, we impose a lag of one month when constructing the growth indicator at each period.

[4] Refer to Amstad and Potter (2009) and Amstad et al. (2014) for more detailed explanations of the UIG measure.

[5] Unfortunately, the UIG measure was unavailable before 1995, so the consumer price index (CPI) measure is used as a proxy normalized to the same UIG when they overlap from January 1995. We also impose a one month lag when constructing our inflation indicator because both the UIG and CPI measures can be obtained with a one month delay.

Identifying economic regimes

Many different econometric techniques are used to model and date economic regimes. For example, Hamilton (1989) develops a Markov-switching model to detect regime shifts, and Chauvet (1998) extends this model to a multivariate framework for estimating the coincident of economic activity and allowing it to switch regimes. Alternatively, Song (2014) proposes a hidden Markov model to estimate and forecast the state of the economy. However, one of the key issues with the Markov-switching model is that parameter estimation has poor out-of-sample robustness and empirical results are highly dependent on the chosen structure of the model. Instead, this study adopts a more practical method to identify economic regimes in a purely data-driven manner. In recent years, several papers, including Vliet and Blitz (2011), Ilmanen et al. (2014), Jurczenko and Teiletche (2018), and Kwon (2022), have also used a similar approach to characterizing economic regimes. Specifically, we define economic regimes as follows:

- Heating Up: Rising growth and rising inflation.
- Goldilocks: Rising growth and falling inflation.
- Slow Growth: Falling growth and falling inflation.
- Stagflation: Falling growth and rising inflation.

To detect regime shifts, we need to extract momentum changes in growth and inflation indicators. To do so, we apply a trend filtering algorithm to each indicator to obtain a trend function by filtering out noise, and identify the momentum changes as sign changes of the slope of the estimated trend function. Specifically, similar to Mulvey and Liu (2016) and Kwon (2022), we employ the l_1 trend filtering method developed by Kim et al. (2009) to acquire the momentum changes in growth and inflation.[6] For the l_1 trend filtering, the trend estimate (x_t) is chosen as the minimizer of the following objective function:

$$\min_{x_t} \left(\frac{1}{2}\right) \sum_{t=1}^{n} \left(y_t - x_t\right)^2 + \lambda \sum_{t=2}^{n-1} \left|x_{t-1} - 2x_t + x_{t+1}\right|, \qquad (1)$$

where y_t is the original time series data (i.e., the growth or inflation indicator in this study) and $\lambda \geq 0$ is a regularization parameter that controls the trade-off between the size of the residual $(y_t - x_t)$ and smoothness of the estimated trend.[7]

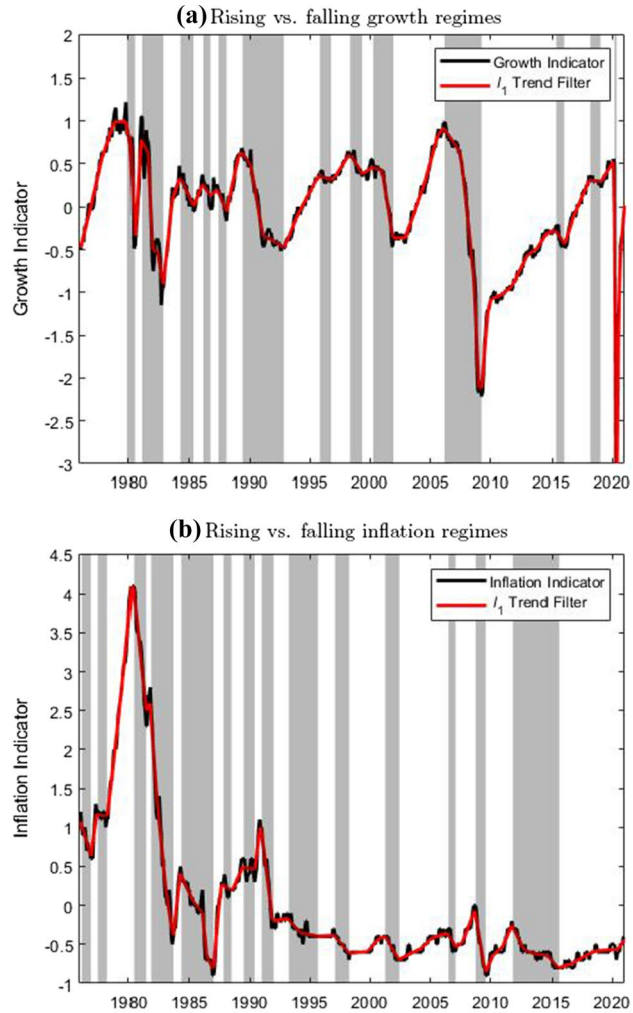


Fig. 3 Two regime plots

The regularization parameter λ is set to a small value of 0.3 to capture sufficient short-term momentum changes in growth and inflation.[8] The l_1 filtering method shares many similar properties with the Hodrick–Prescott (H–P) filtering, but the key difference is that it produces a piecewise linear trend estimate.[9] In other words, the l_1 trend estimate is an affine function over each time interval such that the kink points correspond to changes in the slope of the estimated trend and can be interpreted as changes in the dynamics of the original data.

[6] The l_1 trend filtering method has been widely used in a variety of fields, including economics, finance, geophysics, and machine learning.

[7] The l_1 trend filter converges to the original data as the regularization parameter λ decreases to zero, whereas it becomes smoother and converges to the best affine (straight-line) fit to the original data as λ increases to ∞.

[8] Because the regularization parameter is critical for determining how frequently economic regimes change over time, we will consider different values of the parameter to check the robustness of our main empirical results in sensitivity analysis.

[9] The H–P filtering adopts the sum of squares of the smoothness of the trend estimate in the second part of the objective function in Equation (1) instead of the sum of absolute values.

Table 1 Asset return and risk characteristics from 1976 to 2020

	US stocks	US bonds	Commodities
Panel A: descriptive statistics			
Annualized return (%)	9.97	7.03	3.66
Annualized volatility (%)	15.03	5.17	20.17
Sharpe ratio	0.38	0.53	− 0.03
Panel B: correlation matrix			
US stocks	1.00	0.19	0.25
US bonds	0.19	1.00	− 0.03
Commodities	0.25	− 0.03	1.00
Panel C: diversification metrics			
Average correlation	0.22	0.08	0.11

The one-month T-bill return is used for the Sharpe ratio

Figure 3 presents two-regime plots of the growth and inflation indicators over the past 45 years. Gray areas represent regimes in which growth or inflation momentum is decelerating, whereas light areas represent regimes in which it is accelerating. In the regime plot for the growth indicator, almost all NBER recession periods are classified as growth-falling regimes. Furthermore, several dates that are not designated as the NBER recession periods are detected as growth-falling regimes, such as the 1997 Asian financial crisis and 2015–2016 Chinese stock market turbulence.

Asset class performance across economic regimes

This section discusses the historical performance of different asset classes. Three traditional assets are considered: US stocks, US bonds, and commodities. For their measures, we employ the MSCI USA Total Return Index, Barclays US Aggregate Bond Total Return Index, and S&P GSCI Commodity Total Return Index. All data are monthly and the sample period is January 1976 to December 2020.

Table 1 describes the asset return and risk characteristics of the entire sample. As reported in Panel A, the annualized absolute returns range from 3.66% for commodities to 9.97% for US stocks. Commodities exhibit the highest volatility of 20.17%, which is more than three times greater than the volatility of US bonds. US bonds have the largest Sharpe ratio, which can be mainly attributed to their low volatility compared to the other risky assets. Although commodities have the poorest individual performance, they are somewhat compensated by their diversifying properties: As shown in Panel C, their average correlation with the other assets is 0.11, which is as low as US bonds, but much lower than US stocks. US bonds have the lowest average correlation and are the most attractive asset in terms of risk diversification.

Table 2 Asset return and risk characteristics through economic regimes

	US stocks	US bonds	Commodities
Panel A: heating up			
Annualized return (%)	16.77	4.32	15.41
Annualized volatility (%)	11.07	3.54	17.86
Sharpe ratio	1.27	0.47	0.71
Panel B: goldilocks			
Annualized return (%)	17.19	5.31	0.12
Annualized volatility (%)	12.96	4.63	16.44
Sharpe ratio	0.99	0.19	− 0.26
Panel C: slow growth			
Annualized return (%)	8.18	14.02	− 14.76
Annualized volatility (%)	17.21	6.44	20.98
Sharpe ratio	0.12	1.24	− 0.99
Panel D: stagflation			
Annualized return (%)	− 3.27	7.05	5.03
Annualized volatility (%)	18.29	5.94	24.15
Sharpe ratio	− 0.45	0.35	0.00

The one-month T-bill return is used for the Sharpe ratio

Table 2 reports the historical performance of different asset classes across economic regimes. As expected, each asset has a clear preference for a particular economic environment. For example, US stocks perform well in the regimes of "Heating Up" or "Goldilocks" where growth momentum is improving with corporate margin expansion and profit growth. Conversely, US bonds experience the opposite pattern, performing well in the regimes of "Slow Growth" or "Stagflation," where growth momentum is decreasing. As a result of the inflation-hedging property, commodities generate the highest absolute and risk-adjusted returns in the regimes of "Heating Up" where both growth and inflation momentum are rising. In particular, in the regimes of "Slow Growth" and "Stagflation," all assets tend to become more volatile because of the increased recession risk, suggesting that risky assets such as US stocks and commodities exhibit relatively poor performance, while US bonds deliver better risk-adjusted returns.

To summarize, the identified economic regimes successfully split the time-varying risk and return features of the different asset classes. This evidence clearly supports the potential benefits of regime-based dynamic strategies by tilting exposures toward attractive assets according to changes in the macro-environment.

Out-of-sample analysis

This section investigates the out-of-sample performance of a regime-based dynamic strategy. We first discuss how to construct benchmark and dynamic portfolios and describe the

settings of the out-of-sample experiment. We then compare the empirical results of the regime-based dynamic strategy to those of the static benchmark.

Portfolio construction methods

We use a risk parity portfolio as a benchmark for performance comparison. This is motivated by the following reasons. First, as shown by Qian (2006) and Maillard et al. (2010), the risk parity portfolio is perfectly diversified in terms of risk contributions by allocating the same risk budget to all components in the portfolio. Second, it has been known to be more robust and less concentrated than other risk-based or mean-variance-optimized portfolios. Third, as described by Asness et al. (2012), the risk parity portfolio has historically outperformed market-cap-weighted or mean-variance-optimized portfolios. Lastly, it has been widely adopted as a benchmark portfolio in many pension funds and other institutional investors, as well as the asset management industry.

To construct a dynamic portfolio, we take the Black–Litterman approach similar to the methodology developed by Haesen et al. (2017) and Jurczenko and Teiletche (2018) by embedding an investor's perspective on expected returns into the benchmark portfolio.[10] Specifically, dynamic portfolio construction consists of the following three steps. The first step is to consider the risk parity portfolio as a reference portfolio and to derive an implied view of expected returns Π through reverse portfolio optimization as follows:

$$\Pi = \delta \Sigma \omega_{\mathrm{RP}}, \tag{2}$$

where δ denotes the risk aversion parameter, Σ refers to the covariance matrix of asset returns, and ω_{RP} is the risk parity portfolio. In other words, we can acquire the implied view of expected returns by assuming that the risk parity portfolio (benchmark) is optimal in a mean-variance sense. The risk aversion parameter δ is assumed to be five based on Haesen et al. (2017) and Dichtl et al. (2019), indicating moderate risk aversion. The second step is to establish a posterior view of expected returns $\overline{\mu}$ using a linear combination of the implied view Π and an investor's regime-based active view Q as follows:

$$\overline{\mu} = (1 - \kappa)\Pi + \kappa Q, \tag{3}$$

where $\kappa \in [0, 1]$ is the confidence for an investor's active view Q. The greater the parameter κ, the higher the confidence for an investor's perspective on expected returns. Therefore, the parameter κ captures how actively an investor can execute his or her dynamic strategy and determines the overall level of the posterior view of expected returns. We calibrate the parameter κ (0.09) to target a 2% tracking error relative to the benchmark. The final step is to apply mean-variance optimization (MVO) using the posterior view of expected returns and to obtain the resulting dynamic portfolio with short-selling restrictions.

Experimental settings

The out-of-sample experiment is conducted as follows. The first 30 years of data (1976–2005) are set to initialize the experiment and the remaining periods (2006–2020) are used to implement the out-of-sample testing. To build the benchmark (risk parity portfolio), we consider three traditional assets of US stocks, US bonds, and commodities, and estimate the covariance matrix employing the past 30 years of historical data. The benchmark portfolio is rebalanced every month using the updated covariance matrix with 30 year rolling samples.

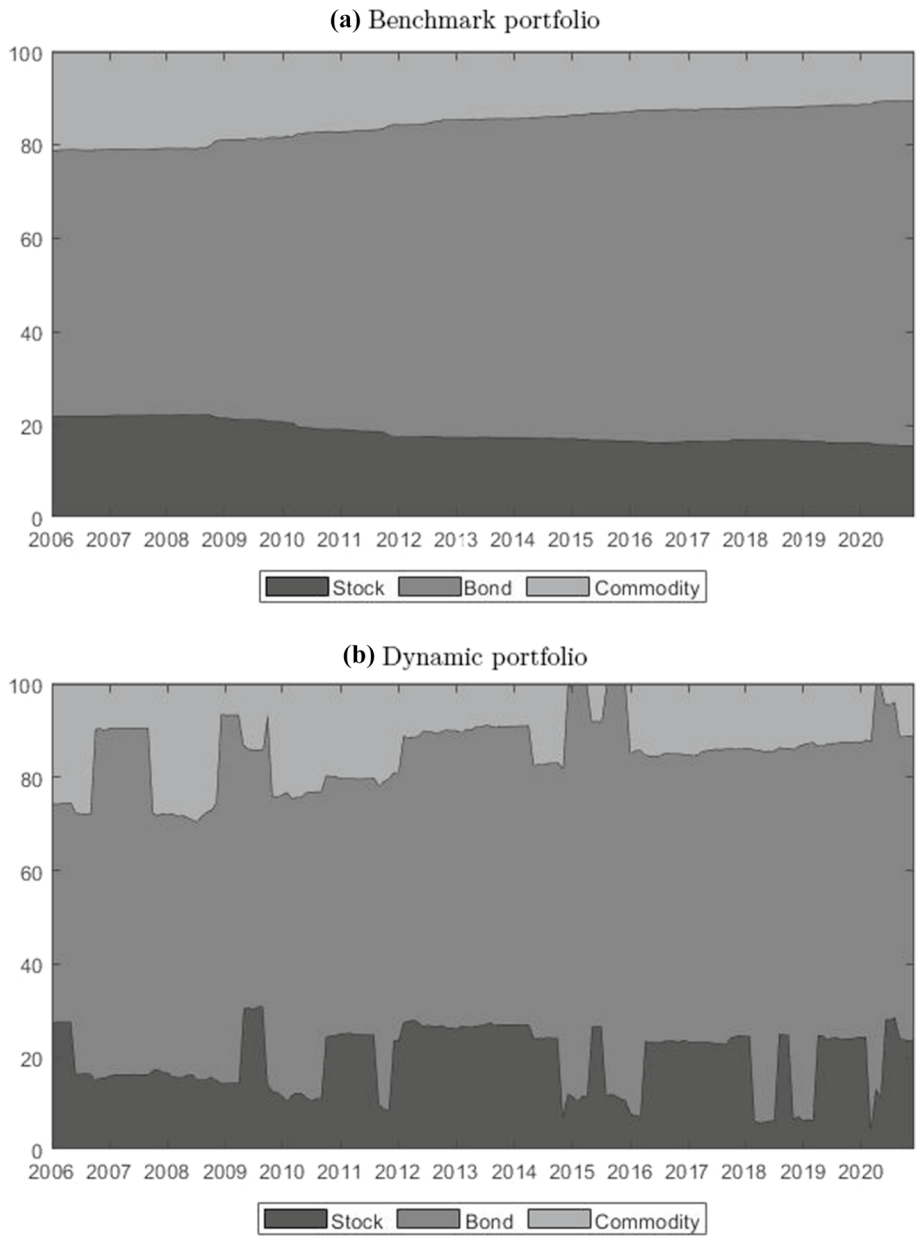
To embed an investor's active view into the benchmark portfolio, we generate the growth and inflation indicators at time t by employing economic data available from the beginning of 1976 to time t and estimate their predicted values at time $t + 1$ by relying on the $AR(1)$ forecasting regression. We then forecast the economic regime at time $t + 1$ by applying the l_1 trend filtering method to each indicator with the predicted values.[11] Based on the regime forecast at time $t + 1$, the dynamic portfolio is rebalanced at the beginning of time $t + 1$ according to an investor's perspective on expected returns for each regime. To be consistent with the covariance estimation, an investor's regime-based perspective on expected returns is assumed to be a simple average return for each regime using the past 30 years of historical samples and the regime-dependent average return is updated every month with new data.[12] This procedure is repeated sequentially by including macroeconomic and financial observations one at a time from January 2006 to December 2020.

[10] As mentioned by Fabozzi et al. (2006) and Jurczenko and Teiletche (2018), the Black–Litterman framework has been extensively employed in the asset management industry when reflecting active views on returns to beat the benchmark. Please refer to Black and Litterman (1991) and Black and Litterman (1992) for additional details on the Black–Litterman framework.

[11] Whenever new economic data are available, we re-normalize the growth and inflation indicators every month and re-estimate their predicted values using the $AR(1)$ rolling regression to forecast the economic regime for the next period.

[12] The use of historical average returns for an investor's regime-based view of expected returns is only adopted for descriptive purposes to test the performance of the regime-based dynamic strategy. In practice, investors may derive their views of expected returns by employ-

Fig. 4 Portfolio weights



Results

We begin by describing how asset weights of the benchmark and dynamic portfolios vary over the entire out-of-sample period. Figure 4 illustrates the results. The benchmark portfolio allocates a large share of US bonds, which can be attributed to their low volatility, as well as to their low correlations with the other two assets. In particular, the weights of the benchmark portfolio are broadly stable, and

Table 3 Out-of-sample performance results

	Benchmark	Dynamic strategy
Annualized return (%)	3.59	5.07
Annualized volatility (%)	6.46	6.60
Sharpe ratio	0.56	0.77
Maximum drawdown (%)	28.99	27.40
Average turnover per month (%)	1.93	5.96
Tracking error (%)	–	2.00
Information ratio	–	0.74
Break-even transaction costs (%)	–	1.95

The one-month T-bill rate is used for the Sharpe ratio

Footnote 12 (continued)

ing a statistical or econometric model to forecast them for each regime.

Table 4 Portfolio performance with different tracking errors

	Benchmark	Dynamic strategy		
		TE = 1%	TE = 2%	TE = 3%
Annualized return (%)	3.59	4.28	5.07	5.63
Annualized volatility (%)	6.46	6.46	6.60	6.97
Sharpe ratio	0.56	0.66	0.77	0.81
Maximum drawdown (%)	28.99	28.21	27.40	27.25
Average turnover per month (%)	1.93	3.50	5.96	8.87
Tracking error (%)	–	1.00	2.00	3.00
Information ratio	–	0.69	0.74	0.68
Break-even transaction costs (%)	–	1.65	1.95	1.65

The one-month T-bill rate is used for the Sharpe ratio

allocations to US bonds tend to increase gradually over time, whereas allocations to US stocks and commodities decrease gradually. In contrast, the weights of the dynamic portfolio fluctuate significantly compared to the benchmark by tilting exposures toward more attractive assets that are expected to exhibit higher returns based on the predicted economic regime.

Table 3 summarizes the out-of-sample performance results. The active approach markedly increases absolute returns by 41% compared to the benchmark. Although the dynamic strategy generates a bit larger volatility than the benchmark, it leads to substantially higher risk-adjusted returns (Sharpe ratio) than the benchmark. Furthermore, given the tracking error target of 2%, the dynamic approach produces a high information ratio of 0.74. To examine whether the dynamic tilt would survive after accounting for trading costs, we adopt the commonly used measure of break-even transaction costs. They are computed as the two-way unitary transaction costs that make the Sharpe ratio of the dynamic portfolio equal to that of the benchmark. As expected, the active strategy results in a larger turnover than the benchmark.[13] Therefore, as long as trading costs are less than approximately 195 bps per two-way transaction, the dynamic strategy generates outperformance net of costs. Considering realistic transaction costs in practice, the dynamic strategy seems to be economically significant. On the whole, we empirically prove that incorporating changes in the macro-environment into dynamic asset allocation strategies is important not only for improving long-term portfolio performance, but also for beating the benchmark.

[13] Following DeMiguel et al. (2009), we define the portfolio turnover as the average sum of the absolute values of the trades across the three available assets. In other words, it captures the average percentage of wealth traded in each period.

Sensitivity analysis

This section assesses the robustness of the empirical results with respect to changes in key parameters and benchmark settings used in the out-of-sample experiment. Overall, we find that the main results largely remain intact, supporting the outperformance of the regime-based dynamic strategy after transaction costs.

Confidence parameter for an investor's view

We first check the sensitivity of dynamic portfolio performance to different values of the confidence parameter κ for an investor's view of expected returns. Recall that the parameter in the baseline experiment is set to 0.09 to match a realistic level of 2% tracking error. Since it determines how actively an investor implements his or her dynamic strategy, it may have a strong impact on the resulting dynamic portfolio, as well as the out-of-sample performance of the dynamic approach.

Table 4 presents the results. Compared to the baseline case of 2% tracking error, the dynamic tilt with 1% tracking error ($\kappa = 0.04$) reduces volatility and average turnover due to the decreased activeness, but it also declines absolute and risk-adjusted returns. In contrast, the dynamic strategy with 3% tracking error ($\kappa = 0.15$) increases not only absolute and risk-adjusted returns, but also volatility and average turnover. On the whole, in any case, the active approach produces higher absolute and risk-adjusted returns than the static benchmark, as well as a high level of information ratio. Most importantly, based on the high level of break-even transaction costs, the regime-based dynamic strategy net of trading costs still seems to be profitable.

Regularization parameter for l_1 trend filtering

Next, we analyze the sensitivity of dynamic portfolio performance to different values of the regularization parameter λ for the l_1 trend filtering. As mentioned previously, this

Table 5 Portfolio performance with different values of the regularization parameter

	Benchmark	Dynamic strategy		
		$\lambda = 0.1$	$\lambda = 0.3$	$\lambda = 0.5$
Annualized return (%)	3.59	5.85	5.07	4.56
Annualized volatility (%)	6.46	6.55	6.60	6.78
Sharpe ratio	0.56	0.89	0.77	0.67
Maximum drawdown (%)	28.99	24.60	27.40	27.40
Average turnover per month (%)	1.93	7.93	5.96	5.22
Tracking error (%)	–	2.12	2.00	1.89
Information ratio	–	1.07	0.74	0.51
Break-even transaction costs (%)	–	2.33	1.95	1.26

The one-month T-bill rate is used for the Sharpe ratio

parameter controls the smoothness of the trend function for the macro-indicators: It converges to the original data as the parameter goes to zero, whereas it becomes smoother and closer to a straight line as the parameter goes to ∞. That is, as the parameter value increases, the trend function becomes smoother so that economic regimes tend to stay in the current state for longer periods. In this regard, the regularization parameter determines how frequently economic regimes switch over time and may have significant effects on the out-of-sample performance of the dynamic strategy.

The results are shown in Table 5. For a smaller value of the regularization parameter ($\lambda = 0.1$), the dynamic approach leads to higher absolute and risk-adjusted returns than the baseline ($\lambda = 0.3$) because of more flexible shifts in economic regimes. On the contrary, for a greater value of the parameter ($\lambda = 0.5$), the dynamic tilt results in lower absolute and risk-adjusted returns than the baseline as economic regimes change more slowly. Nevertheless, in any case, the active strategy generates not only higher absolute and risk-adjusted returns than the static benchmark, but also a high level of information ratio. In particular, considering break-even transaction costs, we can still justify the outperformance of the regime-based dynamic strategy accounting for trading costs.

Risk aversion parameter

This subsection examines how dynamic portfolio performance changes depending on different values of the risk aversion parameter δ. This parameter is used not only when finding the implied view of expected returns from the benchmark portfolio, but also when constructing the dynamic portfolio through mean-variance optimization. In this aspect, the risk aversion parameter may substantially affect the portfolio performance of the dynamic strategy.

Table 6 Portfolio performance with different values of the risk aversion parameter

	Benchmark	Dynamic strategy		
		$\delta = 3$	$\delta = 5$	$\delta = 7$
Annualized return (%)	3.59	5.56	5.07	4.69
Annualized volatility (%)	6.46	7.14	6.60	6.42
Sharpe ratio	0.56	0.78	0.77	0.73
Maximum drawdown (%)	28.99	27.85	27.40	27.40
Average turnover per month (%)	1.93	8.84	5.96	4.64
Tracking error (%)	–	2.98	2.00	1.50
Information ratio	–	0.66	0.74	0.73
Break-even transaction costs (%)	–	1.50	1.95	2.02

The one-month T-bill rate is used for the Sharpe ratio

Table 6 reports the results. The active approach with a greater value of the risk aversion parameter ($\delta = 7$) leads to lower annualized absolute return and volatility than the baseline case ($\delta = 5$). This is because higher risk aversion causes an investor to invest more value into safe assets such as US bonds, rather than risky assets such as US stocks and commodities. As a result, it slightly reduces the Sharpe ratio and information ratio mainly due to the decreased absolute return. Conversely, the dynamic strategy with a smaller value of the risk aversion parameter ($\delta = 3$) increases the absolute return and volatility, inducing an investor to be willing to take risks and execute the active investing more aggressively. Overall, in any case, the dynamic tilt still results in higher absolute and risk-adjusted returns than the static benchmark, as well as a high level of information ratio. In particular, based on the high level of break-even transaction costs, the regime-based dynamic approach still appears to outperform the static benchmark net of trading costs.

Benchmark portfolio

In this subsection, we investigate whether the out-of-sample results are robust with respect to changes in benchmark portfolios. Remember that the risk parity portfolio has been chosen as a benchmark for performance comparison and the implied view of expected returns has been derived from the benchmark through reverse portfolio optimization. To check the sensitivity of this choice, we consider two different benchmark portfolios with all other parameter values remaining the same as in the baseline: One is the equally weighted portfolio and the other is the MVO portfolio.

The results are described in Table 7. Specifically, for the case of the equally weighted benchmark portfolio, the dynamic strategy greatly improves annualized absolute and risk-adjusted returns by more than twice, yielding a quite high level of information ratio of 0.62. More importantly, the calculated break-even transactions costs are large enough

Table 7 Portfolio performance with different benchmarks

	Benchmark: Equal weight	Dynamic strategy	Benchmark: MVO	Dynamic strategy
Annualized return (%)	1.13	2.56	6.60	8.12
Annualized volatility (%)	11.88	11.30	7.54	8.04
Sharpe ratio	0.10	0.23	0.88	1.02
Maximum drawdown (%)	43.94	41.50	27.28	29.58
Average turnover per month (%)	2.54	6.45	3.26	7.82
Tracking error (%)	–	2.32	–	2.44
Information ratio	–	0.62	–	0.65
Break-even transaction costs (%)	–	1.92	–	1.23

The one-month T-bill rate is used for the Sharpe ratio

Table 8 Portfolio performance with different asset classes

	Benchmark: 2 assets	Dynamic strategy	Benchmark: 4 assets	Dynamic strategy
Annualized return (%)	7.63	9.38	3.51	5.45
Annualized volatility (%)	4.53	4.23	7.34	7.32
Sharpe ratio	1.68	2.22	0.48	0.74
Maximum drawdown (%)	15.21	10.96	32.76	29.27
Average turnover per month (%)	1.24	3.96	2.04	6.93
Tracking error (%)	–	1.27	–	1.97
Information ratio	–	1.38	–	0.98
Break-even transaction costs (%)	–	4.75	–	2.34

The one-month T-bill rate is used for the Sharpe ratio

to justify the profitability of the regime-based dynamic approach. For the case of the MVO benchmark portfolio, the dynamic tilt also leads to higher absolute and risk-adjusted returns. However, since the MVO portfolio is very sensitive to changes in expected returns, the active approach results in a relatively high level of average turnover, reducing the break-even transaction costs. Nevertheless, considering realistic trading costs, the dynamic strategy is still found to be profitable after transaction costs. In sum, the outperformance of the regime-based dynamic strategy is broadly robust with regard to changes in benchmark portfolios.

Asset classes

This subsection analyzes the sensitivity of the empirical results with respect to different choices of asset classes. We have used three major asset classes in the out-of-sample exercise: US equities, US bonds, and commodities. To check the robustness of this choice, fixing all other parameter values constant as in the baseline, we consider two different cases: One is the case in which only two assets of US equities and US bonds are used, and the other is the case in which one more asset is added. In the latter case, we include the non-US developed markets equities measured by MSCI EAFE Total Return Index for the following two reasons. One reason is that the US business cycle is closely associated with the other developed countries' economic conditions and financial markets due to trade and financial linkages. The other reason is that the MSCI EAFE Total Return Index provides long enough time series consistent with the original asset classes.

Table 8 shows the portfolio performance with different asset classes. For the two-asset case, the dynamic approach increases the annualized return and Sharpe ratio by 23% and 32%, respectively. In particular, it leads to a significantly high level of information ratio due to the increased return and low tracking error, as well as large break-even transaction costs. The similar results can also be found in the four-asset case. The dynamic tilt increases the annualized return and Sharpe ratio by more than 50%, delivering a high information ratio of 0.98. The break-even transaction costs are also high enough to guarantee the profitability of the dynamic approach net of trading costs. To conclude, the outperformance of the regime-based dynamic strategy mainly remains robust with regard to different asset classes.

Table 9 Portfolio performance with different choices of economic variables

Dynamic strategy						
	All	Excl. yield spread	Excl. credit spread	Excl. jobless claims	Excl. building permits	Excl. VIX
Annualized return (%)	5.07	4.93	5.01	4.95	5.02	4.58
Annualized volatility (%)	6.60	6.90	6.79	7.02	6.76	6.82
Sharpe ratio	0.77	0.71	0.74	0.70	0.74	0.67
Maximum drawdown (%)	27.40	28.87	28.87	33.36	28.87	27.40
Average turnover per month (%)	5.96	6.03	5.37	5.63	6.40	4.31
Tracking error (%)	2.00	1.97	1.74	1.94	2.02	1.93
Information ratio	0.74	0.68	0.81	0.70	0.71	0.51
Break-even transaction costs (%)	1.95	1.52	1.89	1.62	1.65	1.51

The one-month T-bill rate is used for the Sharpe ratio

Choices of economic variables

Finally, we explore how sensitive the out-of-sample results are to our choices of economic variables. Recall that we have employed the five real and financial variables to construct the growth indicator: Yield spread, credit spread, initial jobless claims, building permits, and VIX. To examine the effects of each variable on dynamic portfolio performance, we leave it out one at a time and check the robustness of the empirical results, holding all other parameter values the same as in the baseline.

Table 9 summarizes the results. On the whole, the performance of the dynamic strategy worsens a bit when one of the leading variables is dropped. In particular, when the VIX index is removed, the deterioration in dynamic portfolio performance seems to be the most pronounced, indicating that it plays a significant role in detecting changes in economic regimes. The yield spread and initial jobless claims also appear to be important, as evidenced by the decreased absolute and risk-adjusted returns, while the credit spread and building permits seem to have relatively modest effects on dynamic portfolio performance. Although the relative importance of each variable differs to some degree, the overall empirical results broadly remain intact. In particular, the break-even transaction costs show that the outperformance of the regime-based dynamic strategy net of trading costs is still found to be robust with regard to changes in our choices of economic variables.

Conclusion

This study develops a practical investment framework for dynamic asset allocation strategies based on changes in economic regimes. Out-of-sample analysis proves that the regime-based dynamic strategy generates better portfolio performance than the static benchmark portfolio, demonstrating the significant benefits of the dynamic approach. The empirical results have crucial implications for portfolio managers who seek to beat the benchmark and improve long-term portfolio performance.

Our work would be extended in various directions. The regime-based asset allocation strategy described in this paper can be easily applied to dynamic factor allocations. It also seems worthwhile to incorporate more signals into the investment framework, such as relative valuations, technical analysis, or changes in market regimes, in an effort to improve portfolio performance. Furthermore, it would be meaningful to expand the asset universe in empirical experiments with realistic constraints. In particular, including alternative assets, such as real estate, private equity, or infrastructure, has become critical as a result of their increasing portfolio weights in institutional investors in recent years.

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Declarations

Conflict of interest All authors declare that we have no conflicts of interest.

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